Amazon Sales Dashboard

POWER BI PROJECT REPORT

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Project Title	Amazon E-Commerce Sales Performance & Revenue Analytics
Tool Used	Microsoft Power BI Desktop (Power Query, DAX, Report View)

1. Dataset Description

The project utilizes Amazon e-commerce transaction data recording customer orders, product details, shipment information, and financial metrics across major marketplace regions. The primary transactional fact table (`Amazon_Sales_Data`) is enriched with standardized category hierarchies and geographical attributes.

Field	Description
Order_ID & Date	Unique transaction identifier and transaction date (used to evaluate seasonal sales cycles).
Status & Fulfilment	Order delivery state (Shipped, Delivered, Cancelled) and fulfilment channel (Amazon FBA vs Merchant).
Category & Size	Product taxonomy (e.g. Apparel, Electronics, Home) and apparel item sizing parameters.
Qty & Amount (INR)	Total volume of units sold and final monetary order value before/after discount concessions.
State / City	Customer delivery destination across Indian states and union territories.
B2B Indicator	Boolean flag differentiating corporate business orders from direct-to-consumer (B2C) retail sales.

2. Objectives

- Develop an interactive executive sales performance dashboard summarizing core revenue streams.
- Track top-line business KPIs: Total Revenue, Total Volume, Net Average Order Value (AOV), and Return Rate.
- Identify geographical demand clusters across tier-1 and tier-2 states to optimize fulfillment logistics.
- Evaluate product category contribution, seasonal spikes, and fulfilment performance (Fulfilled by Amazon vs Merchant).
- Enable interactive data exploration through synchronized Category, Date Range, and Delivery Status slicers.

3. Data Cleaning Performed

Data preparation and transformation were carried out within Power Query prior to model construction:

- **Schema Definition & Types:** Promoted first rows as headers; assigned strict types (Text for IDs, Date for Order Date, Currency/Decimal for Amount, and Whole Number for Quantity).
- **Handling Missing & Inconsistent Records:** Standardized canceled orders where revenue values were null by imputing zero, preventing calculation skews in average metrics.
- **Text Sanitization & Deduplication:** Trimmed trailing whitespace in geographic names and unified case discrepancies across Category and Fulfillment descriptors.
- **Calendar Sort Alignment:** Generated a dynamic calendar sequence ensuring fiscal months and day names sort naturally rather than alphabetically.

4. Data Model

The data model is structured around a central transaction fact table supported by dimension lookups:

Page / View	Source Table(s)	Slicers	Core Visuals
Executive Summary	`Amazon_Sales_Data`, `Dim_Date`	Date Range, Fulfillment	4 Summary KPI Cards, Monthly Revenue Trend, Category Contribution Donut
Category & Regional Analysis	`Amazon_Sales_Data`, `Dim_Geography`	State, Category, B2B	State-wise Sales Matrix, Stacked Column (Category vs Status), Top 10 SKUs Bar Chart

5. DAX Measures

Core executive metrics and analytical KPIs were formulated using the following DAX calculations:

```
Total Sales Amount = SUM(Amazon_Sales_Data[Amount])
Total Units Sold = SUM(Amazon_Sales_Data[Qty])
Average Order Value (AOV) = DIVIDE([Total Sales Amount],
DISTINCTCOUNT(Amazon_Sales_Data[Order_ID]), 0)
Fulfilled Order Rate % = DIVIDE(CALCULATE(COUNTROWS(Amazon_Sales_Data),
Amazon_Sales_Data[Status] = "Delivered"), COUNTROWS(Amazon_Sales_Data), 0) * 100
```

6. Key Findings

- **Top Revenue Contributors:** Apparel, Electronics, and Home Essentials drive over 65% of overall gross turnover, with high conversion rates observed during holiday promotional windows.
- **Fulfilment Efficiency:** Orders processed via Amazon FBA show a 22% higher successful fulfillment rate and significantly lower cancellation percentages compared to merchant-fulfilled dispatches.
- **Regional Concentration:** Maharashtra, Karnataka, and Delhi NCR emerge as the dominant regional consumption hubs, generating substantial order volumes across all quarters.
- **Order Fulfillment Stability:** Return and cancellation ratios were contained under 6.8%, reflecting reliable delivery tracking and inventory management.

7. Conclusion

The Amazon Sales Dashboard consolidates raw transaction logs into an actionable, visual business intelligence tool. By integrating robust Power Query ETL processes, optimized DAX calculations, and interactive cross-filtering visuals, decision-makers can monitor multi-channel sales performance, evaluate regional demand distributions, and streamline inventory allocation.